Postoperative Epidural Fibrosis: Challenges and Opportunities - A Review

Guido Lewik[1)], Gerrit Lewik[2)], Lena S. Müller[2)], Alexander von Glinski[1)], Tobias L. Schulte[1)] and Tobias Lange[1)]

1) Department of Orthopedics and Trauma Surgery, Katholisches Klinikum Bochum - St. Josef Hospital, Ruhr University Bochum, Bochum, Germany
2) Department of General and Trauma Surgery, BG University Hospital Bergmannsheil, Ruhr University Bochum, Bochum, Germany

Abstract:

Postoperative epidural fibrosis (EF) is still a major limitation to the success of spine surgery. Fibrotic adhesions in the epidural space, initiated via local trauma and inflammation, can induce difficult-to-treat pain and constitute the main cause of failed back surgery syndrome, which not uncommonly requires operative revision.

Manifold agents and methods have been tested for EF relief in order to mitigate this longstanding health burden and its socioeconomic consequences. Although several promising strategies could be identified, few have thus far overcome the high translational hurdle, and there has been little change in standard clinical practice. Nonetheless, notable research progress in the field has put new exciting avenues on the horizon.

In this review, we outline the etiology and pathogenesis of EF, portray its clinical and surgical presentation, and critically appraise current efforts and novel approaches toward enhanced prevention and treatment.

Keywords:

Epidural Fibrosis, Peridural Fibrosis, Epidural Adhesions, Peridural Adhesions, Laminectomy, Failed Back Surgery Syndrome, Epiduroscopy, Epidural Adhesiolysis, Epidurolysis

Spine Surg Relat Res 2024; 8(2): 133-142
dx.doi.org/10.22603/ssrr.2023-0106

Introduction

Postoperative epidural fibrosis (EF) is a highly prevalent complication of spine surgery and the most common cause of failed back surgery (pain) syndrome (FBSS)[1)].

EF is elicited by surgical trauma-induced inflammation, leading to expansive extracellular matrix synthesis, which can result in leg and back pain-provoking adhesions to the dura and nerve roots.

Indeed, the presence of extensive EF is associated with a 3.2-fold increase in persistent postdiscectomy low back pain[2)] and symptomatic EF affects up to 30% of patients following lumbar laminectomy[3)]. EF further necessitated an operative revision in 4%-9% of postdiscectomy cases[4)]. Revision surgery in patients with EF, however, holds an increased risk of intraoperative complications such as bleeding, nerve root injury, and dural laceration[5)].

In the clinic, symptomatic EF is typically diagnosed by correlating symptoms with patient history and magnetic resonance imaging (MRI) findings. Nevertheless, MRI was found to be approximately five times less likely to detect high-grade EF in patients at risk when compared to endoscopic assessment (16.1% vs. 91.0%, respectively)[6)], which however is neither widely available nor a standard spine center technique.

Indeed, the intraoperative appearance and texture of fibrotic tissue help inform this diagnosis, yet the technical complexity of detachment differs vastly and is hardly predictable with routine preoperative planning. This heterogeneity remains a conundrum hindering more selective indication for revision surgery because data on intraoperative features as well as tissue analysis of human EF samples are lacking and thus cannot be matched to prior imaging or symptoms.

In preclinical research utilizing animal models, conversely, EF evaluation centers around macroscopic and histological analyses, whereas MRI or pain and sensitivity scores are

Corresponding author: Guido Lewik, Guido.Lewik@ruhr-uni-bochum.de
Received: July 28, 2023, Accepted: July 30, 2023, Advance Publication: September 4, 2023
Copyright © 2024 The Japanese Society for Spine Surgery and Related Research

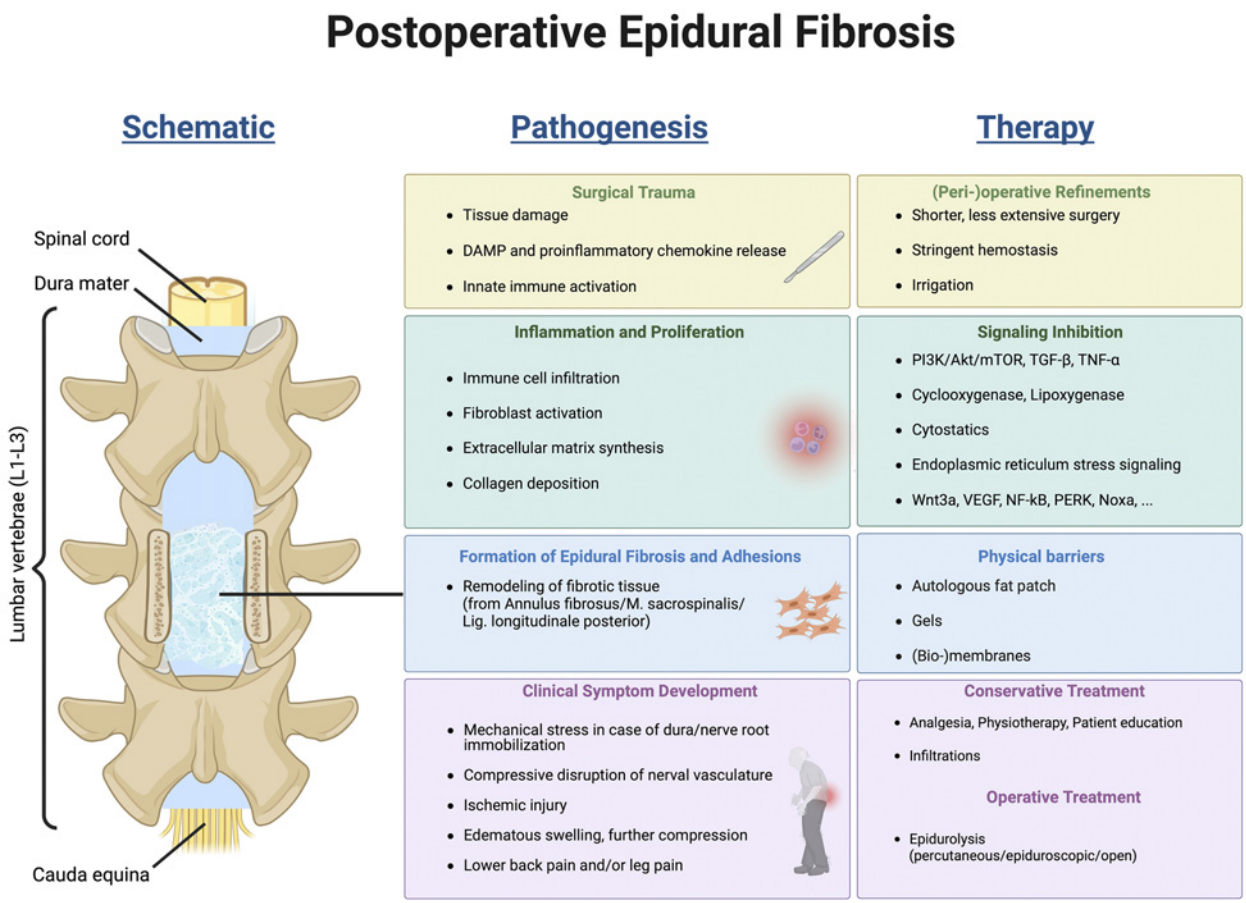

Figure 1. Summary of postoperative EF development, presentation, and treatment. *Created with BioRender.com.*

rarely employed. This discrepancy may contribute to difficulties in translational efforts toward EF-targeted therapies.

As preventive measures to alleviate EF formation, a plethora of substances has been tested preclinically, predominantly aiming to dampen local inflammation and/or create a physical barrier to contain dural adhesion[7]. However, of the many promising candidates, few have progressed to clinical trials, and none have demonstrated reliable patient benefit.

Besides these challenges, the traditional causal treatment of EF is operative fibrotic tissue removal, which holds an increased risk of intraoperative complications and, cynically, replicates its trigger mechanism. Lacking well-established methods, less invasive approaches have therefore been probed to remedy EF-evoked symptoms.

Being a longstanding impediment in spine surgery, research on EF has spanned almost five decades[8] and intensified in recent years, while an unmet need for effective strategies persists.

This review is intended to comprehensively present relevant knowledge regarding EF pathogenesis, past and current efforts toward preventive measures and the translational hurdle they face, as well as novel treatment strategies (Fig. 1). It aims to highlight challenges and opportunities for clinical translation and provide guidance in the collective endeavor to overcome this tenacious restriction of spinal surgery success.

EF Etiology and Pathogenesis

Understanding the etiology and pathogenesis of EF is critical to develop effective preventive measures.

Archetypal postoperative scar tissue formation follows a three-stage process. It commences with 3-5 days of early inflammatory reaction encompassing hemostasis, coagulation, and chemokine-driven immune cell infiltration and activation. Subsequently, 2-3 weeks of fibrous transformation arises from fibroblast-driven extracellular matrix production in response to cytokines including transforming growth factor (TGF)-β1, interleukin (IL)-6, and fibroblast growth factor in the postinjury environment. Finally, tissue remodeling takes effect over months to years[9].

The specific origin of postoperative tissue adhesions in the epidural space has been debated from as early as 1948 when Key and Ford made out the posterior aspect of the annulus fibrosus (AF) as the primary lesion for postoperative adhesions extending to the nerve roots[10]. However, LaRocca and Macnab proposed in 1974 that scarring may primarily derive from the sacrospinalis muscle surface adjacent to the dural surface, in the process of forming "the laminectomy membrane"[11]. Other authors have later attributed EF formation to both sacrospinalis and AF tissue as well as the posterior longitudinal ligament[12].

The principal pathogenic function of EF tissue is compressive disruption of the neural microvasculature, leading to

ischemic injury and thereby reinforcing compression via edematous swelling[13,14]. When EF causes immobilization of neural structures, mechanical stress like extension represents an additional source of discomfort and pain[13].

Although it has been argued that EF represents granulation tissue and may be considered normal wound healing[15], it is nowadays widely accepted as non-physiologic[14].

Importantly, antagonizing inflammatory processes does not appear to impair local physiological remodeling such as intervertebral disk (IVD) matrix production[16], whereas inflammation triggers AF degradation[17] and fibrotic AF changes correlate with IVD herniation progression[18].

Accordingly, normal tissue regeneration is understood to occur alongside scarring, which has inspired interventions utilizing inflammation control, biomaterial-bridging between resection margins, and regenerative cell therapy application to avoid leaving the field to detrimental processes[19].

Overall, there is a lack of mechanistic information on postsurgical tissue development in the epidural sphere, yet site-specific tissue microenvironments and immune responses are known to affect cytomolecular pathways, resulting in a variety of prospective therapeutic targets.

While the underlying mechanisms need to be clarified further, the role of several pathways in EF pathogenesis could be inferred from studies trialing preventive strategies (Table 1)[7].

Clinical and Surgical EF Presentation

Daily position changes cause movement of nerval structures, which likely provokes and aggravates pain sensitizations in patients with EF. While there is little evidence on physiological dural motion, cadaver investigations have shown lumbosacral nerve root displacement of up to 5 mm during passive straight leg raise[20]. Spinal flexion is anticipated to result in even greater motions of the lumbar dural sac, putting significantly more mechanical stress on EF-adhered nerval structures.

Preoperative MRI helps to determine the extent of EF, and epiduroscopy has lately been trialed as an alternative method, which demonstrates even greater sensitivity than MRI for low-level fibrosis[21]. Nonetheless, any type of visualization currently fails to adequately predict the complexity of surgical removal. Intraoperatively found EF tissue adherence to surrounding structures, which primarily drives the risk of complications, exhibits striking heterogeneity. Studies on human EF tissue samples could expand our understanding of epidural tissue formation and the underlying mechanisms responsible for its variable development and presentation.

Preventive Strategies to Reduce EF Development

The primary research focus of EF therapy lies in prevention. Preventive approaches are mainly built on antiproliferative agents, biomaterials creating mechanical barriers with optional drug release functionality, and surgical refinements[7].

Antiproliferative and anti-inflammatory agents trialed preclinically have successfully targeted the profibrotic PI3K/Akt/mTOR axis, which is promoted by TGF-β, as well as NF-kB, Wnt3, VEGF signaling and several more (Table 1). Novel studies have even presented promising results in laminectomized rats with postsurgery radial extracorporeal shock wave therapy[22] and hyperbaric oxygen treatment[23].

Barrier agents

Another mechanism used to prevent epidural adhesions is physical barrier creation. Autologous fat patches represent a popular, inexpensive method, yet its effectiveness is debated and rare, albeit serious, complications such as acute cauda equina syndrome have been reported[63,64]. While compression injuries could occur using any kind of mechanical barrier, autologous fat may also have limited long-time effects due to swift atrophy, which is in line with one study that shows no postoperative benefit after a mean 2.6-year follow-up in patients that underwent IVD herniation repair[65]. The repertoire has therefore been expanded to gels and (bio-)membranes, which are likewise absorbable and hold an added risk of unfavorable immune responses, but offer more flexible composition and shaping as well as combination with drug or cell loading strategies for enhanced therapeutic effect.

Following laminectomy *in vivo*, preclinical studies have shown promising EF reduction employing such distancing agents to separate the dura from overlying structures[66] (Table 2).

Especially, the use of scaffold-seeded adipose-derived stem cells or mesenchymal stem cell-derived exosomes may offer great advantages by uniting early, temporary distancing with anti-inflammatory signaling and regenerative capabilities[67,68].

The specific features of materials employed, which are beyond the scope of this article, have recently been reviewed in greater detail by Wang et al[7].

Clinical translation and (peri-)operative strategies

Few preclinically tested strategies have thus far made their way to clinical application despite eager efforts over the last three decades. Most prominently, the bioabsorbable polymer gel Adcon-L had reached the clinical stage owing to reported success in pain and EF reduction before conflicting results and adverse events such as subdural hematoma, spontaneous intracranial hypotension, and impaired tissue healing ultimately led to market withdrawal[76-78].

Unfortunately, the anticipated advantages of alternative biomaterials over fat autografts have hitherto not convincingly materialized in increased patient benefit. Two more recent studies indeed reported that fat autografts displayed better results at postoperative clinical and MRI scoring when compared to Gelfoam in patients with IVD herniation[79] and laminectomy[80], respectively. Locally applied mitomycin C,

Table 1. Overview of Preclinical Trials with Antiproliferative or Anti-inflammatory Agents to Prevent EF Formation.

Agent(s) used	Pathway	Effect	Model/species	Year	Citation
Decorin-soaked spongostan	PI3K/AKT/mTOR and Smad2/3	Antiproliferative effects on fibroblast cultures *in vitro* and epidural fibrosis/adhesions following laminectomy *in vivo*	Laminectomy/rats	2021	24
Pirfenidone	PI3K/AKT/mTOR	Reduced EF, fibroblast proliferation, migration, and adhesion	Laminectomy/rats	2021	25
Thymoquinone	Cyclooxygenase and lipoxygenase	Decrease in EF and increase in new bone and capillary volume	Laminectomy/rats	2021	26
Curcumin	Cyclooxygenase-2 and lipoxygenase	Reduced EF, inflammation, and medulla spinalis retraction	Laminectomy/rats	2021	27
Laminin α5	PI3K/AKT/mTOR	Inhibited fibroblast proliferation *in vitro*; Laminin α5 expression was associated with EF development *in vivo*	Laminectomy/rats	2020	28
DNAse I	DNA degradation and cleavage of neutrophil extracellular traps	Reduced EF	Laminectomy/Mice	2020	29
Triptolide	PI3K/AKT/mTOR	Reduced surgery-induced EF	Laminectomy/rats	2019	30
Artesunate	Autophagy-mediated p53/ $p21^{waf1/cip1}$ pathway	Reduced EF and fibroblast proliferation	Laminectomy/rats	2019	31
Emodin	PERK signaling	Reduced EF, increased fibroblast apoptosis	Laminectomy/rats	2019	32
Apigenin	Wnt3a/β-catenin	Inhibition of EF and fibroblast proliferation	Laminectomy/rats	2019	33
Methyl palmitate	Cyclooxygenase-2	Reduced EF and inflammatory cell densities	Laminectomy/rats	2018	34
Etanercept	TNF-α	Reduced EF with greater effect upon systemic than local administration	Laminectomy/rats	2014, 2018	35, 36
Homoharringtonine	PI3K/AKT/mTOR and endoplasmic reticulum stress signaling	Inhibition of fibroblast proliferation and induction of fibroblast apoptosis *in vitro*, suppression of post-laminectomy EF *in vivo*	Laminectomy/rats	2017	37
Methotrexate	Endoplasmic reticulum stress signaling	Reduced EF, hydroxyproline content, and fibroblast numbers	Laminectomy/rats	2017	38
Tacrolimus	miR-429 and RhoE	Reduced EF and promoted fibroblast apoptosis	Laminectomy/rats	2017	39
MMC(-controlled release membranes)	Cytostasis	Reduced EF and hydroxyproline concentrations	Laminectomy/rats	2006, 2017	40, 41
Rapamycin	mTOR	Reduced fibroblast proliferation and EF *in vivo*	Laminectomy/rats	2016	42
Suramin	Growth factor inhibition	Reduced fibroblast proliferation *in vitro* and EF *in vivo*	Laminectomy/rats	2016	43
Povidone-iodine, rifampicin, and hydrogen peroxide	Antimicrobial	Povidone-iodine and hydrogen peroxide reduced EF *in vivo*, whereas rifampin did not	Laminectomy/rats	2016	44
Salvianolic acid B	Proposedly ROS, TNF-α, and IL-1β	Inhibition of EF, fibroblast proliferation, vascularization, and inflammation	Laminectomy/rats	2014, 2016	45, 46
CCN5	Smad6-CCN2	Inhibition of profibrotic fibroblast development	Laminectomy/rats	2015	47
Dexamethasone-soaked spongostan	Proposedly via VEGF/ VEGFR2	Reduced postoperative EF	Laminectomy/rats	2015	48
Rosuvastatin	TGF-β1	Reduced EF with greater effect upon systemic than local administration	Laminectomy/rats	2015	49
Licofelone	Cyclooxygenase and lipoxygenase	Reduced EF, hydroxyproline deposits, and inflammatory factor expression	Laminectomy/rats	2015	50
MMC	miR-200b and RhoE	Reduced epidural scar hyperplasia, increased fibroblast apoptosis, and autophagy	Laminectomy/rats	2015	51
all-trans retinoic acid	NF-kB signaling	Reduced EF, hydroxyproline content, fibroblast, and inflammatory cell density	Laminectomy/rats	2014	52
ERK2 (siRNAs)	Downregulation of collagen deposition, IL-6, and TGF-β1	Reduced EF, collagen expression, and inflammation	Laminectomy/rats	2014	53

Table 1. Overview of Preclinical Trials with Antiproliferative or Anti-inflammatory Agents to Prevent EF Formation. (continued)

Agent(s) used	Pathway	Effect	Model/species	Year	Citation
Verapamil	TGF-β1, IL-6	Reduced EF, fibroblast proliferation, TGF-β1, and IL-6 expression	Laminectomy/rats	2014	54
Parecoxib-soaked absorbable gelatin sponge	Cyclooxygenase-2	Reduced fibroblast and inflammatory cell densities, as well as fibrous adherence	Laminectomy/rats	2013	55
Hydroxycamptothecin	NOXA upregulation	Reduced EF, fibroblasts, and hydroxyproline concentration	Laminectomy/rabbits	2011, 2013	56, 57
Tacrolimus	TGF-β1/SMAD	Reduced epidural scar thickness	Laminectomy/rats	2011	58
Pimecrolimus	Cyclooxygenase-2, NFAT	Reduced dural thickness and adherence	Laminectomy/rats	2009	59
Aceclofenac	Cyclooxygenase	Reduced thickness of peridural adhesions and inflammatory cell counts within	Laminectomy/rabbits	2008	60
IFN-γ	TGF-β1	Reduced EF, fibroblast and inflammatory cell density	Laminectomy/rats	2008	61
Mitomycin C (MMC), 5-fluorouracil (5-FU), and cyclosporin A (CsA)	Cytostasis, cytostasis, and immunosuppression	Reduced fibroblasts and inflammatory cells as well as postoperative EF in MMC and 5-FU groups but not CsA	Laminectomy/rats	2007	62

another auspicious strategy, did also not achieve clinical benefit despite notably reduced EF in MRI scoring postdiscectomy[81].

Nevertheless, further perioperative strategies should not be overlooked when aiming for EF control.

Minimal remaining blood within the epidural space has been described to alleviate EF formation[82], whereas hematoma presence appears to aggravate it[83]. Larger surgical procedures could also be connected to more abundant EF[2]. Hence, spine surgery patients may benefit from stringent hemostasis and shorter operating times, possibly due to less extensive surgery.

Similarly, irrigation of the epidural site is a common yet unstandardized step in surgery that may hold great potential for optimization. Kizilay and colleagues have reported significant EF reduction upon irrigation with 10% povidone-iodine solution (PVP-I) in laminectomized rats[44]; however, the ideal concentration and duration of application are still unknown. It appears desirable to explore the ideal equilibrium between PVP-I dose- and time-dependent bactericidal and cytotoxic effects in this complex healing process[84,85].

Treatment Strategies (Conservative, Interventional, and Operative)

When patients display symptomatic EF, conservative measures are initiated. If refractory to conservative treatment, patients can be offered interventional or surgical release.

Conservative treatment of symptomatic EF encompasses patient education, physiotherapy, analgesia with non-steroidal anti-inflammatory drugs, and injections of corticosteroid and local anesthetic agents[86,87]. Moreover, as recently reviewed, spinal cord stimulation has produced encouraging results in patients with FBSS[88].

EF symptoms refractory to conservative therapy can subsequently be addressed using two key approaches, namely fluid administration and mechanical release.

Fluid administration, such as via epidural injections, appears to reduce pain as a function of the volume used, prompting the hypothesis that cytokine dilution and dispersion may be crucial for its success[89,90]. Additives to injections, such as steroids, have shown rather discouraging results[91], whereas a small pilot study with the application of ozone reported moderate success[92].

Mechanical release of fibrotic tissue requires epidural adhesiolysis, termed epidurolysis, and can be done via percutaneous, endoscopic, or open surgical access.

Percutaneous epidurolysis has been used since the early 1980s[93] and features catheter entry into the epidural space with subsequent application of saline or pharmacological agents to ease adhesions. By removing or diluting proinflammatory cytokines, the volume of fluid injected appears to be the primary factor in pain alleviation[90]. Access can be gained from ventral or dorsal, with ventral access showing slightly improved visual analog scale scores up to 6 months postoperatively[94]. Epidural application of steroids remains debated, however, as evidence shows non-superiority at 1, 6, and 12 months postsurgery when compared to intravenous application[95]. The minimally invasive approach of percutaneous epidurolysis appears to also come at the cost of an increased risk for bleeding, dural puncture, abscesses, and meningitis[96].

In endoscopic epidurolysis, termed epiduroscopy, lytic agents are applied to the site of adhesions with visual guidance through a spinal endoscope[89]. This more novel approach was developed to increase accuracy and thereby efficacy and, indeed, has demonstrated more than 50% chronic back pain relief in the majority of patients suffering from EF in various studies[97-101]. A current meta-analysis of epidu-

Table 2. Overview of Preclinical Trials with Distancing/barrier Agents to Prevent EF Formation.

Agent(s) used	Assessment and effect	Model/species	Year	Citation
•Polyurethane (PU) membrane •Expanded polytetrafluoroethylene (ePTFE) membrane •Bacterial cellulose (BC) membrane •BC membrane+human umbilical cord MSC exosomes	MRI (12 weeks) and macroscopic/histological analysis (1 year): Reduced epidural fibrosis and peridural adhesions.	Laminectomy/ rabbits	2018	68
•Cova (membrane barrier) •Tisseel (fibrin sealant) •Adcon-L gel	Histology (6 weeks): All agents reduced peridural fibrosis.	Laminectomy/rats	2017	69
•Floseal (hemostatic matrix) •Ostene (alkylene oxide copolymer) •Adcon-L	Histology (6 weeks): All agents reduced peridural fibrosis.	Laminectomy/rats	2017	70
•Gelatin sponge with dexamethasone	Macroscopic/histological analysis (4, 8, and 12 weeks): No obvious adhesion formation and reduced epidural VEGF and VEGFR2 expression upon combination treatment, whereas gelatin sponge or dexamethasone did not prevent EF.	Laminectomy/rats	2015	48
•Poly-L-glutamic acid/chitosan (PLGA/CS) barrier	MRI (12 and 24 weeks): Less epidural fibrosis, scar tissue, peridural adhesion, foreign body reaction, and lower pressure of spinal cord. Macroscopic/histological analysis (24 weeks): Less scar tissue, less epidural adhesion, and lower fibroblast density.	Laminectomy/ rabbits	2014	71
•Mitomycin C with PEG film •Mitomycin C with PLGA film	Histology (4 weeks): Prevented epidural scarring and adhesions.	Laminectomy/rats	2008, 2014	72, 73
•Poly (lactic-co-glycolic acid) (PLGA) scaffold with or without adipose-derived stem cells (ASCs)	MRI (after 1, 12, and 24 weeks): Coarse, high-density scar tissue in PLGA group at 12 weeks; continuous linear adipose tissue regenerated along spinal cord at 24 weeks. Macroscopic/histological analysis (24 weeks): Distinct area of adipose tissue overlying dura in cell-scaffold complex treated group.	Laminectomy/ rabbits	2012	67
•Freeze-dried amniotic membrane (FAM) •Cross-linked amniotic membrane (CAM) •Autologous free fat (AFF)	Macroscopic/histological analysis (1, 6, and 12 weeks): Less scar tissue and adhesion tenacity in CAM group vs. FAM and non-treated. Reduced fibroblast infiltration and EF in CAM, similar to AFF group.	Laminectomy/ dogs	2009	66
•Hyaluronan gel •Bioabsorbable macropore sheet	Macroscopic/histological analysis (1, 3, and 8 weeks): Both reduced EF; Hyaluronan gel decreased inflammatory cell migration.	Laminectomy/rats	2005, 2006	74, 75

roscopic EF release in FBSS patients has attested clinically relevant reduction in pain and disability scores after 6-12 months[102]. Still, there is a lack of randomized controlled trials that compare percutaneous and endoscopic epidurolysis, and recent systematic reviews conclude that more research is needed to confirm these preliminary findings[103,104].

Open surgical adhesiolysis is carried out in patients who are refractory to less-invasive treatment methods or where revision surgery is scheduled for other reasons than, or additional reasons to, clinically relevant EF. To reduce repeated EF triggering by surgical trauma, one novel strategy is non-cutting dissection using MESNA in the Chemically Assisted DISSection (CADISS) System, for which recent results in the first 19 patients showed operative success without complications[105]. Additional studies are required to assess the longer-term results of CADISS use for EF release.

Similarly, more data are required to back up the report of improved clinical outcomes upon preoperative low-dose external-beam irradiation before revision surgery in 10 patients with symptomatic postdiscectomy EF[106].

The evidence providing a rationale for epidurolysis has recently been thoroughly reviewed by Urits et al.[14], who highlighted both safety and efficacy but stressed the need for further well-designed studies.

Conclusion

The widely present complications of EF and epidural adhesions result in substantial morbidity for patients undergoing spinal canal surgery. Although early research has pro-

vided directive insight into its etiology and pathogenesis, manifold preventive approaches have remained without successful clinical translation. Interventional, less-invasive treatment methods have made remarkable progress in alleviating EF-evoked pain and novel approaches could help optimize (peri-)operative therapy toward both prevention and release of adhesions.

Nevertheless, efforts to utilize novel biomolecular methods for EF tissue characterization should be increased to facilitate causative treatment development. Additionally, more high-quality clinical studies are warranted to probe and expedite the clinical translation of promising strategies in the endeavor to root out EF from the spinal surgery complication conspectus.

Conflicts of Interest: The authors declare that there are no relevant conflicts of interest.

Sources of Funding: None.

Author Contributions: Gu.L., T.L., and T.S. conceived and planned the review with additions from Ge.L., L.M., and A.v.G.; Gu.L. wrote the manuscript with contributions from Ge.L., L.M., and A.v.G.; T.L. and T.S. helped prepare the manuscript with comments and pieces of advice. All authors approved the final manuscript and agreed to be accountable for all aspects of the work ensuring that questions related to the accuracy or integrity of any part of the work are properly investigated and resolved.

References

1. Zeinalizadeh M, Miri SM, Ardalan FA, et al. Reduction of epidural fibrosis and dural adhesions after lamina reconstruction by absorbable cement: an experimental study. Spine J. 2014;14(1):113-8.
2. Ross JS, Robertson JT, Frederickson RC, et al. Association between peridural scar and recurrent radicular pain after lumbar discectomy: magnetic resonance evaluation. ADCON-L European study group. Neurosurgery. 1996;38(4):855-61; discussion 61-3.
3. Jönsson B, Strömqvist B. Repeat decompression of lumbar nerve roots. A prospective two-year evaluation. J Bone Joint Surg Br. 1993;75(6):894-7.
4. Guyer RD, Patterson M, Ohnmeiss DD. Failed back surgery syndrome: diagnostic evaluation. J Am Acad Orthop Surg. 2006;14(9):534-43.
5. de Tribolet N, Porchet F, Lutz TW, et al. Clinical assessment of a novel antiadhesion barrier gel: prospective, randomized, multicenter, clinical trial of ADCON-L to inhibit postoperative peridural fibrosis and related symptoms after lumbar discectomy. Am J Orthop. 1998;27(2):111-20.
6. Bosscher HA, Heavner JE. Incidence and severity of epidural fibrosis after back surgery: an endoscopic study. Pain Pract. 2010;10(1):18-24.
7. Wang H, Sun W, Fu D, et al. Update on biomaterials for prevention of epidural adhesion after lumbar laminectomy. J Orthop Translat. 2018;13:41-9.
8. Langenskiöld A, Kiviluoto O. Prevention of epidural scar formation after operations on the lumbar spine by means of free fat transplants. A preliminary report. Clin Orthop Relat Res. 1976;115(115):92-5.
9. Sun Y, Wang L, Sun S, et al. The effect of 10-hydroxycamptothecine in preventing fibroblast proliferation and epidural scar adhesion after laminectomy in rats. Eur J Pharmacol. 2008;593(1-3):44-8.
10. Key JA, Ford LT. Experimental intervertebral-disc lesions. J Bone Joint Surg Am. 1948;30A(3):621-30.
11. LaRocca H, Macnab I. The laminectomy membrane. Studies in its evolution, characteristics, effects and prophylaxis in dogs. J Bone Joint Surg Br. 1974;56B(3):545-50.
12. Songer MN, Ghosh L, Spencer DL. Effects of sodium hyaluronate on peridural fibrosis after lumbar laminotomy and discectomy. Spine. 1990;15(6):550-4.
13. Rydevik B, Lundborg G, Bagge U. Effects of graded compression on intraneural blood blow. An in vivo study on rabbit tibial nerve. J Hand Surg Am. 1981;6(1):3-12.
14. Urits I, Schwartz RH, Brinkman J, et al. An evidence based review of epidurolysis for the management of epidural adhesions. Psychopharmacol Bull. 2020;50(4 Suppl 1):74-90.
15. Long DM. Failed back surgery syndrome. Neurosurg Clin N Am. 1991;2(4):899-919.
16. Teixeira GQ, Leite Pereira C, Castro F, et al. Anti-inflammatory chitosan/poly-γ-glutamic acid nanoparticles control inflammation while remodeling extracellular matrix in degenerated intervertebral disc. Acta Biomater. 2016;42:168-79.
17. Saggese T, Teixeira GQ, Wade K, et al. Georg Schmorl Prize of the German Spine Society (DWG) 2018: combined inflammatory and mechanical stress weakens the annulus fibrosus: evidences from a loaded bovine AF organ culture. Eur Spine J. 2019;28(5):922-33.
18. Castro AL, Ribeiro-Machado C, Oliveira CM, et al. Fibrotic alterations in human annulus fibrosus correlate with progression of intervertebral disc herniation. Arthritis Res Ther. 2022;24(1):25.
19. Atala A, Irvine DJ, Moses M, et al. Wound healing versus regeneration: role of the tissue environment in regenerative medicine. MRS Bull. 2010;35(8):597-606.
20. Bellier A, Latreche A, Tissot L, et al. Movements of the lumbosacral nerve roots in the spinal canal induced by straight leg raising test: an anatomical study. Surg Radiol Anat. 2018;40(11):1223-30.
21. Guner D, Asik I, Ozgencil GE, et al. The correlation of epidural fibrosis with epiduroscopic and radiologic imaging for chronic pain after back surgery. Pain Phys. 2021;24(8):E1219-26-e26.
22. Haberal B, Şimşek EK, Akpınar K, et al. Impact of radial extracorporeal shock wave therapy in post-laminectomy epidural fibrosis in a rat model. Jt Dis Relat Surg. 2021;32(1):162-9.
23. Lv P, Zhao J, Su W, et al. An experimental novel study: hyperbaric oxygen treatment on reduction of epidural fibrosis via down-regulation of collagen deposition, IL-6, and TGF-β1. Eur J Orthop Surg Traumatol. 2015;25 Suppl 1:S53-8.
24. Ding Q, Wei Q, Sheng G, et al. The preventive effect of decorin on epidural fibrosis and epidural adhesions after laminectomy. Front Pharmacol. 2021;12:774316.
25. Zhang X, Zhang J, Liu Y, et al. Pirfenidone inhibits fibroblast proliferation, migration or adhesion and reduces epidural fibrosis in rats via the PI3K/AKT signaling pathway. Biochem Biophys Res Commun. 2021;547:183-91.
26. Türköz D, Türköz A, Önger ME, et al. Stereological and histopathological evaluation of the effect of thymoquinone on peridural fibrosis following laminectomy in rats. Turk J Med Sci. 2021;51(1):375-81.

27. Demirel C, Turkoz D, Yazicioglu IM, et al. The preventive effect of curcumin on the experimental rat epidural fibrosis model. World Neurosurg. 2021;145:e141-8.

28. Liu P, Chen H, Yan L, et al. Laminin α5 modulates fibroblast proliferation in epidural fibrosis through the PI3K/AKT/mTOR signaling pathway. Mol Med Rep. 2020;21(3):1491-500.

29. Jin Z, Sun J, Song Z, et al. Neutrophil extracellular traps promote scar formation in post-epidural fibrosis. npj Regen Med. 2020;5(1):19.

30. Dai J, Sun Y, Chen D, et al. Negative regulation of PI3K/AKT/mTOR axis regulates fibroblast proliferation, apoptosis and autophagy play a vital role in triptolide-induced epidural fibrosis reduction. Eur J Pharmacol. 2019;864:172724.

31. Wan Q, Chen H, Li X, et al. Artesunate inhibits fibroblasts proliferation and reduces surgery-induced epidural fibrosis via the autophagy-mediated p53/p21(waf1/cip1) pathway. Eur J Pharmacol. 2019;842:197-207.

32. Xiong G, Chen H, Wan Q, et al. Emodin promotes fibroblast apoptosis and prevents epidural fibrosis through PERK pathway in rats. J Orthop Surg Res. 2019;14(1):319.

33. Jiao R, Chen H, Wan Q, et al. Apigenin inhibits fibroblast proliferation and reduces epidural fibrosis by regulating Wnt3a/β-catenin signaling pathway. J Orthop Surg Res. 2019;14(1):258.

34. Kizilay ZAP, Kahraman Cetin NAP. Effect of methyl palmitate on the formation of epidural fibrosis in an experimental epidural fibrosis model. J Invest Surg. 2018;31(6):469-74.

35. Turkoglu E, Tuncer C, Dinc C, et al. The effect of etanercept on spinal epidural fibrosis in a postlaminectomy rat model. Turk Neurosurg. 2014;24(4):506-11.

36. Kizilay Z, Cetin NK, Topcu A, et al. Effect of etanercept on the formation of epidural fibrosis in an experimental model. Turk Neurosurg. 2018;28(1):111-7.

37. Li X, Wang S, Dai J, et al. Homoharringtonine prevents surgery-induced epidural fibrosis through endoplasmic reticulum stress signaling pathway. Eur J Pharmacol. 2017;815:437-45.

38. Chen H, Yan L, Wang J, et al. Methotrexate prevents epidural fibrosis through endoplasmic reticulum stress signalling pathway. Eur J Pharmacol. 2017;796:131-8.

39. Li X, Chen H, Wang S, et al. Tacrolimus induces fibroblasts apoptosis and reduces epidural fibrosis by regulating miR-429 and its target of RhoE. Biochem Biophys Res Commun. 2017;490(4):1197-204.

40. Lee JY, Stenzel W, Impekoven P, et al. The effect of Mitomycin C in reducing epidural fibrosis after lumbar laminectomy in rats. J Neurosurg Spine. 2006;5(1):53-60.

41. Xie H, Wang B, Shen X, et al. MMC controlled-release membranes attenuate epidural scar formation in rat models after laminectomy. Mol Med Rep. 2017;15(6):4162-8.

42. Sun Y, Zhao S, Li X, et al. Local application of rapamycin reduces epidural fibrosis after laminectomy via inhibiting fibroblast proliferation and prompting apoptosis. J Orthop Surg Res. 2016;11(1):58.

43. Dai J, Li X, Yan L, et al. The effect of suramin on inhibiting fibroblast proliferation and preventing epidural fibrosis after laminectomy in rats. J Orthop Surg Res. 2016;11(1):108.

44. Kizilay Z, Cetin NK, İsmailoglu Ö, et al. The effects of rifampin, povidone-iodine and hydrogen peroxide on the formation of epidural fibrosis in the experimental epidural fibrosis model. Inflammation. 2016;39(4):1495-502.

45. Chen F, Zuo Z, Wang K, et al. Study on salvianolic acid B in the reduction of epidural fibrosis in laminectomy rats. BMC Musculoskelet Disord. 2014;15:337.

46. Chen F, Wang C, Sun J, et al. Salvianolic acid B reduced the formation of epidural fibrosis in an experimental rat model. J Orthop Surg Res. 2016;11(1):141.

47. Xu H, Liu C, Sun Z, et al. CCN5 attenuates profibrotic phenotypes of fibroblasts through the Smad6-CCN2 pathway: potential role in epidural fibrosis. Int J Mol Med. 2015;36(1):123-9.

48. Tian F, Dou C, Qi S, et al. Preventive effect of dexamethasone gelatin sponge on the lumbosacral epidural adhesion. Int J Clin Exp Med. 2015;8(4):5478-84.

49. Gürer B, Kahveci R, Gökçe EC, et al. Evaluation of topical application and systemic administration of rosuvastatin in preventing epidural fibrosis in rats. Spine J. 2015;15(3):522-9.

50. Zhang K, Zhao J, Su W, et al. Immunomodulatory effectiveness of licofelone in preventing epidural fibrosis in post-laminectomy rat. Eur J Orthop Surg Traumatol. 2015;25(Suppl 1):S63-8.

51. Sun Y, Ge Y, Fu Y, et al. Mitomycin C induces fibroblasts apoptosis and reduces epidural fibrosis by regulating miR-200b and its targeting of RhoE. Eur J Pharmacol. 2015;765:198-208.

52. Zhang C, Kong X, Ning G, et al. All-trans retinoic acid prevents epidural fibrosis through NF-κB signaling pathway in post-laminectomy rats. Neuropharmacology. 2014;79:275-81.

53. Zhang C, Kong X, Liu C, et al. ERK2 small interfering RNAs prevent epidural fibrosis via the efficient inhibition of collagen expression and inflammation in laminectomy rats. Biochem Biophys Res Commun. 2014;444(3):395-400.

54. Wang Z, Wang Y, Xie P, et al. Calcium channel blockers in reduction of epidural fibrosis and dural adhesions in laminectomy rats. Eur J Orthop Surg Traumatol. 2014;24(Suppl 1):S293-8.

55. Sae-Jung S, Jirarattanaphochai K. Prevention of peridural fibrosis using a cyclooxygenase-2 inhibitor (nonsteroidal anti-inflammatory drug) soaked in absorbable gelatin sponge: an experimental comparative animal model. Spine (Phila Pa 1976). 2013;38(16):E985-91.

56. Zhu L, Ni B, Liu J, et al. Hydroxycamptothecin liposomes inhibit collagen secretion and induce fibroblast apoptosis in a postlaminectomy rabbit model. Eur J Orthop Surg Traumatol. 2013;23(Suppl 1):S85-91.

57. Yang J, Ni B, Liu J, et al. Application of liposome-encapsulated hydroxycamptothecin in the prevention of epidural scar formation in New Zealand white rabbits. Spine J. 2011;11(3):218-23.

58. Ismailoglu O, Albayrak B, Gulsen I, et al. Topical application of tacrolimus prevents epidural fibrosis in a rat postlaminectomy model: histopathological and ultrastructural analysis. Turk Neurosurg. 2011;21(4):630-3.

59. Cemil B, Tun K, Kaptanoglu E, et al. Use of pimecrolimus to prevent epidural fibrosis in a postlaminectomy rat model. J Neurosurg Spine. 2009;11(6):758-63.

60. Sandoval MA, Hernandez-Vaquero D. Preventing peridural fibrosis with nonsteroidal anti-inflammatory drugs. Eur Spine J. 2008;17(3):451-5.

61. Emmez H, Kardes O, Dogulu F, et al. Role of antifibrotic cytokine interferon-gamma in the prevention of postlaminectomy peridural fibrosis in rats. Neurosurgery. 2008;62(6):1351-7; discussion 7-8.

62. Yildiz KH, Gezen F, Is M, et al. Mitomycin C, 5-fluorouracil, and cyclosporin A prevent epidural fibrosis in an experimental laminectomy model. Eur Spine J. 2007;16(9):1525-30.

63. Imran Y, Halim Y. Acute cauda equina syndrome secondary to free fat graft following spinal decompression. Singapore Med J. 2005;46(1):25-7.

64. Mayer PJ, Jacobsen FS. Cauda equina syndrome after surgical treatment of lumbar spinal stenosis with application of free auto-

genous fat graft. A report of two cases. J Bone Joint Surg Am. 1989;71(7):1090-3.

65. Görgülü A, Simşek O, Cobanoğlu S, et al. The effect of epidural free fat graft on the outcome of lumbar disc surgery. Neurosurg Rev. 2004;27(3):181-4.

66. Tao H, Fan H. Implantation of amniotic membrane to reduce postlaminectomy epidural adhesions. Eur Spine J. 2009;18(8): 1202-12.

67. Xu J, Chen Y, Yue Y, et al. Reconstruction of epidural fat with engineered adipose tissue from adipose derived stem cells and PLGA in the rabbit dorsal laminectomy model. Biomaterials. 2012;33(29):6965-73.

68. Wang B, Li P, Shangguan L, et al. A novel bacterial cellulose membrane immobilized with human umbilical cord mesenchymal stem cells-derived exosome prevents epidural fibrosis. Int J Nanomedicine. 2018;13:5257-73.

69. Kazanci A, Gurcan O, Gurcay AG, et al. Effects of topical Cova™, Tisseel® and Adcon® gel application on the development of spinal peridural fibrosis: an experimental study in rats. Turk Neurosurg. 2017;27(6):962-8.

70. Gurcan O, Gurcay AG, Kazanci A, et al. Is the use of hemostatic matrix (FloSeal) and alkylene oxide copolymer (Ostene) safe in spinal laminectomies? Peridural fibrosis assessment. Acta Orthop Traumatol Turc. 2017;51(2):165-8.

71. Li C, Wang H, Liu H, et al. The prevention effect of poly (L-glutamic acid)/chitosan on spinal epidural fibrosis and peridural adhesion in the post-laminectomy rabbit model. Eur Spine J. 2014;23(11):2423-31.

72. Kurt G, Aytar MH, Doğulu F, et al. A comparison of the local effectiveness of Mitomycin C, aprotinin, and Adcon-L in experimental peridural fibrosis. Surg Neurol. 2008;70(6):608-13; discussion 13.

73. Sui T, Zhang J, Du S, et al. Potential risk of mitomycin C at high concentrations on peripheral nerve structure. Neural Regen Res. 2014;9(8):821-7.

74. Akeson WH, Massie JB, Huang B, et al. Topical high-molecular-weight hyaluronan and a roofing barrier sheet equally inhibit postlaminectomy fibrosis. Spine J. 2005;5(2):180-90.

75. Schimizzi AL, Massie JB, Murphy M, et al. High-molecular-weight hyaluronan inhibits macrophage proliferation and cytokine release in the early wound of a preclinical postlaminectomy rat model. Spine J. 2006;6(5):550-6.

76. Ivanic GM, Pink PT, Schneider F, et al. Prevention of epidural scarring after microdiscectomy: a randomized clinical trial comparing gel and expanded polytetrafluoroethylene membrane. Eur Spine J. 2006;15(9):1360-6.

77. Kasimcan MO, Bakar B, Aktaş S, et al. Effectiveness of the biophysical barriers on the peridural fibrosis of a postlaminectomy rat model: an experimental research. Injury. 2011;42(8):778-81.

78. Kim SB, Lim YJ. Delayed detected unexpected complication of ADCON-L® gel in lumbar surgery. J Korean Neurosurg Soc. 2010;48(3):268-71.

79. Rajpal K, Singh J, Bahadur R, et al. Postoperative epidural fibrosis prevention: which is better-autologous fat versus gelfoam. Asian Spine J. 2022;16(3):343-51.

80. Sobti S, Grover A, John BPS, et al. Prospective randomized comparative study to evaluate epidural fibrosis and surgical outcome in patients undergoing lumbar laminectomy with epidural autologous free fat graft or gelfoam: a preliminary study. Int J Appl Basic Med Res. 2018;8(2):71-5.

81. Liu L, Sui T, Hong X, et al. Inhibition of epidural fibrosis after microendoscopic discectomy with topical application of Mitomycin C: a randomized, controlled, double-blind trial. J Neurosurg Spine. 2013;18(5):421-7.

82. Weinstein JN, Lurie JD, Olson PR, et al. United States' trends and regional variations in lumbar spine surgery: 1992-2003. Spine. 2006;31(23):2707-14.

83. Mohi Eldin MM, Abdel Razek NM. Epidural fibrosis after lumbar disc surgery: prevention and outcome evaluation. Asian Spine J. 2015;9(3):370-85.

84. Berkelman RL, Holland BW, Anderson RL. Increased bactericidal activity of dilute preparations of povidone-iodine solutions. J Clin Microbiol. 1982;15(4):635-9.

85. Lemans JVC, Öner FC, Wijdicks SPJ, et al. The efficacy of intrawound vancomycin powder and povidone-iodine irrigation to prevent surgical site infections in complex instrumented spine surgery. Spine J. 2019;19(10):1648-56.

86. Zav'yalov DM, Peretechikov AV. Prevention and treatment of postoperative epidural scar adhesions. Zh Vopr Neirokhir Im N N Burdenko. 2016;80(6):115-7. rus.

87. Bush K, Hillier S. A controlled study of caudal epidural injections of triamcinolone plus procaine for the management of intractable sciatica. Spine. 1991;16(5):572-5.

88. Kurt E, Noordhof RK, van Dongen R, et al. Spinal cord stimulation in failed back surgery syndrome: an integrative review of quantitative and qualitative studies. Neuromodulation. 2022;25 (5):657-70.

89. Lee F, Jamison DE, Hurley RW, et al. Epidural lysis of adhesions. Korean J Pain. 2014;27(1):3-15.

90. Rabinovitch DL, Peliowski A, Furlan AD. Influence of lumbar epidural injection volume on pain relief for radicular leg pain and/or low back pain. Spine J. 2009;9(6):509-17.

91. Häckel M, Masopust V, Bojar M, et al. The epidural steroids in the prevention of epidural fibrosis: MRI and clinical findings. Neuro Endocrinol Lett. 2009;30(1):51-5.

92. Magalhães FN, Soares SC, Torres JM, et al. Effects of ozone applied by spinal endoscopy in patients with chronic pain related to failed back surgery syndrome: a pilot study. Neuropsychiatr Dis Treat. 2013;9:1759-66.

93. Racz GB, Sabonghy M, Gintautas J, et al. Intractable pain therapy using a new epidural catheter. JAMA. 1982;248(5):579-81.

94. Oh CH, Ji GY, Cho PG, et al. The catheter tip position and effects of percutaneous epidural neuroplasty in patients with lumbar disc disease during 6-months of follow-up. Pain Phys. 2014; 17(5):E599-608.

95. Hu A, Gu X, Guan X, et al. Epidural versus intravenous steroids application following percutaneous endoscopic lumbar discectomy. Med. 2018;97(18):e0654.

96. Talu GK, Erdine S. Complications of epidural neuroplasty: a retrospective evaluation. Neuromodulation. 2003;6(4):237-47.

97. Vymazal J, Kříž R. Vertebroplasty and epiduroscopy as seen by interventional radiologist. Cas Lek Cesk. 2018;157(4):203-7.

98. Helm S, Hayek SM, Colson J, et al. Spinal endoscopic adhesiolysis in post lumbar surgery syndrome: an update of assessment of the evidence. Pain Phys. 2013;16(2 Suppl):SE125-50.

99. Pereira P, Severo M, Monteiro P, et al. Results of lumbar endoscopic adhesiolysis using a radiofrequency catheter in patients with postoperative fibrosis and persistent or recurrent symptoms after discectomy. Pain Pract. 2016;16(1):67-79.

100. Manchikanti L, Boswell MV, Rivera JJ, et al. [ISRCTN 16558617] A randomized, controlled trial of spinal endoscopic adhesiolysis in chronic refractory low back and lower extremity pain. BMC Anesthesiol. 2005;5:10.

101. Donato AD, Fontana C, Pinto R, et al. The effectiveness of en-

doscopic epidurolysis in treatment of degenerative chronic low back pain: a prospective analysis and follow-up at 48 months. Acta Neurochir Suppl. 2011;108:67-73.

102. Geudeke MW, Krediet AC, Bilecen S, et al. Effectiveness of epiduroscopy for patients with failed back surgery syndrome: a systematic review and meta-analysis. Pain Pract. 2021;21(4): 468-81.

103. Helm S, 2nd, Racz GB, Gerdesmeyer L, et al. Percutaneous and endoscopic adhesiolysis in managing low back and lower extremity pain: a systematic review and meta-analysis. Pain Phys. 2016;19(2):E245-82.

104. Brito-García N, García-Pérez L, Kovacs FM, et al. Efficacy, effectiveness, safety, and cost-effectiveness of epidural adhesiolysis for treating failed back surgery syndrome. A systematic review. Pain Med. 2019;20(4):692-706.

105. Makhchoune M, Collard X, Triffaux M, et al. The utility of the CADISS® system in the dissection of epidural fibrosis in revision lumbar spine surgery (A case series). Ann Med Surg. 2022; 83:104718.

106. Gerszten PC, Moossy JJ, Flickinger JC, et al. Low-dose radiotherapy for the inhibition of peridural fibrosis after reexploratory nerve root decompression for postlaminectomy syndrome. J Neurosurg. 2003;99(3 Suppl):271-7.

Spine Surgery and Related Research is an Open Access journal distributed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License. To view the details of this license, please visit (https://creativecommons.org/licenses/by-nc-nd/4.0/).